

Conditional Obstructions and Launch Renormalization in Accelerated Collatz Dynamics

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August 2026

Abstract

For the accelerated odd Collatz map $\mathcal{C}(m) = (3m + 1)/2^{v_2(3m+1)}$, we study the least positive integer $r(D)$ realizing a prescribed finite valuation word D , and its relation to the exponential-scale problem left open in the residue-axis reformulation of the Effective Occupation Conjecture. We prove that the exact 2-adic transport state at any index where the coarse residue has stabilized to $r(D)$ collapses to the single expression $T_j = -m_j^* 3^{-j}$, where m_j^* is the j -th accelerated iterate of $r(D)$; consequently, the letters following such an index are exactly the accelerated valuation word of the orbit of m_j^* .

We use this *launch-state collapse* to show that a word with anomalously small realizer decomposes into a short free prefix followed by a long forced suffix equal to an orbit segment of a smaller integer. We analyze three possible structural classes for this suffix: bounded-period periodic behavior, controlled by a fixed rational 2-adic anchor and the correspondence between integral periodic anchors and genuine Collatz cycles; cycle-riding behavior, controlled unconditionally and sharply for the trivial cycle but remaining conditional for hypothetical nontrivial cycles; and an irregular residual class. For the periodic branch we obtain an ε -dependent terminal-repetition fraction bound

$$1 - (1 - \varepsilon) \log_3 2 + o(1),$$

whose frontier limit as $\varepsilon \rightarrow 0$ is $1 - \log_3 2 \approx 0.36907$.

For the irregular branch we prove a renormalization/residual result: the forced suffix is itself an accelerated orbit segment of a smaller integer, but its inherited confinement corridor grows linearly in the original word length rather than remaining fixed. Thus launch renormalization preserves the form of the single-window problem without reducing it to a uniformly fixed-corridor problem. We identify the corridor-uniform lifetime hypothesis that would suffice to close this residual branch and derive the corresponding explicit exponential least-realizer threshold, while labeling that hypothesis as open.

The analysis throughout concerns a single contiguous confined orbit segment. It does not bound the number of separated returns to the confinement region, does not establish a bound on the multi-window occupation quantity O_c , and therefore does not prove the full Effective Occupation Conjecture. No proof of the Collatz conjecture is claimed. The contribution is a rigorous decomposition of the single-window least-realizer problem into proved and conditional obstruction mechanisms, together with an exact identification of the renormalization-stable residual class that prevents closure by the present finite structural methods.

1 Introduction

The Collatz (or $3x + 1$) conjecture concerns the iteration of the map $T(n) = n/2$ for n even, $T(n) = (3n + 1)/2$ for n odd. Its *accelerated* form, folding each odd-to-odd transition into a single

step,

$$\mathcal{C}(m) = \frac{3m+1}{2^{v_2(3m+1)}}, \quad m \text{ odd},$$

has been the object of a substantial literature relating its orbits to 2-adic and combinatorial structure on parity or valuation sequences [15, 9, 2, 12, 16]. A finite orbit segment of length N is encoded by its *valuation word* $D = (d_1, \dots, d_N)$, $d_i = v_2(3m_{i-1} + 1)$, and the central quantity of interest here is $r(D)$, the least positive odd integer whose orbit realizes D exactly.

This note is a self-contained record of a conditional-obstruction program aimed at bounding $r(D)$ from below for words satisfying a fixed drift corridor (“confinement”), a problem shown in prior work by the author to be essentially equivalent to a residue-axis reformulation of the Effective Occupation Conjecture (EOC) for the accelerated map [4]. The program produced a coherent structural result even though it does not close the underlying conjecture: confined words with an anomalously small realizer decompose, via an exact 2-adic identity, into a short free prefix and a long forced suffix that is literally the orbit of a smaller positive integer. Two of the three structural classes into which that forced suffix falls admit genuine, independently sourced obstructions; the third does not, and we prove precisely why it cannot, by finite means, be reduced to the first two.

No proof of the Collatz conjecture is claimed anywhere in this note. The contribution is a rigorous decomposition of a least-realizer/confinement problem into conditional obstruction mechanisms and a renormalization-stable residual class, together with the exact quantitative form of the theorem that would be needed to close that residual class, audited so that its dependence on all relevant parameters is explicit rather than left to informal description.

Summary of results and their status

Throughout we use the epistemic labels **K** (known result, cited at point of use), **B** (background, established in the author’s own prior technical notes and restated here for self-containedness), **N** (new result proved in this note), and **C** (computational observation). The main new results are:

- the launch-state collapse and forced-continuation equivalence (Theorem 5.4), showing the forced suffix of a pinned word is exactly the accelerated orbit of an explicit integer;
- the terminal-suffix depth identity (Theorem 5.3) with sharp, non-asymptotic bounds;
- the anchor–cycle dichotomy (Proposition 7.1), adapting the classical Böhm–Sontacchi correspondence [3] to the accelerated-word anchor and connecting it explicitly to the periodic realizer floor of [5];
- the periodic-tail and cycle-tail conditional obstructions (§7, §8), assembled from these together with the Eliahou–Roziar drift identity;
- the Renormalization/Residual Proposition (Proposition 9.1), stating precisely in what sense the irregular residual class regenerates the original problem;
- the corridor-scaling correction (§11), showing that the natural closing hypothesis must be uniform over a corridor width that grows *linearly* in the word length, not a fixed constant, which is a strictly stronger requirement than the fixed-corridor form of the original problem;
- the conditional implication (Proposition 11.3) that this correctly quantified hypothesis, together with the two proved obstructions, is sufficient to yield an exponential least-realizer floor.

2 Relation to previous $3x + 1$ work

The correspondence between finite parity or valuation sequences and 2-adic residue classes of realizing integers originates with Terras [15] and Everett [9], who established the basic combinatorial and modular structure of stopping-time behavior for the $3x + 1$ map. Bernstein and Lagarias [2] made this correspondence precise as a genuine topological conjugacy: the map $\Phi : \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ conjugating the 2-adic shift to the (accelerated) Collatz map, so that a finite parity/valuation prefix corresponds exactly to a cylinder set in $\mathbb{Z}_2$. The exact-realizer machinery used below is the same correspondence specialized to *positive integer* representatives of a finite cylinder, together with an explicit terminal-parity refinement of the residue class (Proposition 4.3 below), which is developed and proved in the author’s companion technical note [7] and restated here as background.

The relation $2^{S_j}m_j = 3^j m_0 + C_j(D)$ used throughout (Definition 3.1, eq. (1)) is the standard telescoped form of the odd-to-odd recursion; the cyclic special case, where $m_0 = m_N$, is exactly the identity used by Böhm and Sontacchi [3] to parametrize candidate Collatz cycles by binary (there) or valuation (here) admissibility vectors, and by Eliahou [8] to derive lower bounds on nontrivial cycle length from Diophantine approximation of $\log_2 3$. We use both papers’ mechanism directly in §7–8. Rozier’s Lower Bound Hypothesis [12] is the entropy-scale conjecture that the residue-axis problem discussed here is a pointwise (as opposed to almost-all) shadow of; we do not use or assume it. The current best unconditional bounds excluding short nontrivial cycles are due to Simons and de Weger [13] and Hercher [10], and the current computational verification range for the conjecture itself is due to Barina [1]; both are used only as external inputs bounding the unknown-cycle branch of §8, never assumed away. Lagarias’s survey [11] and Tao’s almost-all result [14] are standard background; nothing in this note relies on Tao’s theorem, since our concern is exceptional (pointwise), not almost-all, behavior.

On the word “renormalization.” Wirsching’s monograph [16] is the standard systematic treatment of $3n + 1$ predecessor sets via 2-adic and measure-theoretic methods; its table of contents (Introduction; Some ideas around $3n + 1$ iterations; Analysis of the Collatz graph; 3-adic averages of counting functions; An asymptotically homogeneous Markov chain; Mixing and predecessor density) and index do not, to the extent we were able to verify, contain a chapter or index entry devoted to a technical construction named “renormalization.” We did not obtain full access to the monograph’s interior pages and therefore cannot certify the absence of the term in running prose; we record explicitly that we searched for a conflicting formal usage and found none, and we use “launch renormalization” below only in the specific, self-contained operational sense of Definition 5.6 and Proposition 9.1 — the statement that the forced continuation of a pinned word is literally the accelerated orbit of a smaller integer, subject to a corridor of explicitly computed width. No identity with any construction in [16] or elsewhere is claimed.

3 The accelerated map and valuation-word notation

Definition 3.1 (Valuation word). *Let m_0 be a positive odd integer and $m_i = \mathcal{C}(m_{i-1})$ for $i \geq 1$. The length- N valuation word of the orbit is $D = (d_1, \dots, d_N)$, $d_i = v_2(3m_{i-1} + 1) \geq 1$. For $0 \leq j \leq N$ write $S_j = \sum_{i=1}^j d_i$ (so $S_0 = 0$), and define the carry sum*

$$C_j(D) = \sum_{i=0}^{j-1} 3^{j-1-i} 2^{S_i}, \quad C_0 = 0,$$

which satisfies the one-step recursion $C_{j+1} = 3C_j + 2^{S_j}$, immediate from re-indexing the defining sum. A positive odd integer m realizes D (or its length- j prefix D_j) if $v_2(3m_{i-1} + 1) = d_i$ for $i = 1, \dots, j$ along its own orbit.

Lemma 3.2 (Affine transport identity; standard). *For every $0 \leq j \leq N$ and every m_0 realizing D ,*

$$2^{S_j} m_j = 3^j m_0 + C_j(D). \quad (1)$$

Proof. Induction on j ; the base case is trivial, and the step follows from $2^{d_j} m_j = 3m_{j-1} + 1$ multiplied by $2^{S_{j-1}}$ and combined with the inductive hypothesis at $j-1$, using the recursion for C_j . $\square$

This is the standard telescoped odd-to-odd recursion underlying the Terras/Everett parity-vector correspondence [15, 9] and its 2-adic conjugacy form [2]; the cyclic case $m_0 = m_N$ is exactly the equation solved by Böhm and Sontacchi [3] to determine candidate cycles.

4 Affine and 2-adic transport identities

Definition 4.1. *For $j \geq 1$ define $\xi_j := -C_j(D) \cdot 3^{-j} \in \mathbb{Z}_2$ (well defined since 3 is a unit in $\mathbb{Z}_2$), and write*

$$\xi_j = \chi_j + 2^{S_j} T_j, \quad \chi_j \in \{0, 1, \dots, 2^{S_j} - 1\}, \quad T_j \in \mathbb{Z}_2.$$

Lemma 4.2 (Oddness; background). *For every $j \geq 1$, $C_j(D)$ is odd, hence ξ_j is odd, hence $1 \leq \chi_j < 2^{S_j}$ and χ_j is odd.*

Proof. $C_j = 3^{j-1} + \sum_{i=1}^{j-1} 3^{j-1-i} 2^{S_i}$; the first term is odd and every later term is even since $S_i \geq 1$ for $i \geq 1$. Hence C_j is odd, and since 3^{-j} is also odd in $\mathbb{Z}_2$, so is $\xi_j = -C_j 3^{-j}$; the reduction of an odd 2-adic integer modulo any 2^k ($k \geq 1$) is odd, in particular nonzero. $\square$

Proposition 4.3 (Exact realizer congruence; background [7]). *The exact realizers of D (of length N) form a single residue class modulo 2^{S_N+1} , and the least positive representative is*

$$r(D) = \chi_N + (1 - \beta_N) 2^{S_N}, \quad (2)$$

where $\beta_N \in \{0, 1\}$ is the bit of ξ_N at position S_N (equivalently, $\beta_N \equiv T_N \pmod{2}$).

Proof sketch. By Lemma 3.2, an integer m_0 satisfying the aggregate congruence $2^{S_N} \mid 3^N m_0 + C_N$ has all formal iterates $m_1, \dots, m_N$ integral, by downward induction on the recursion (each step dividing by a smaller power of 2 than is guaranteed by the hypothesis at the next index, using that 3 is a unit modulo any power of 2). For $j \leq N-2$ both m_j, m_{j+1} are then forced odd and the valuation $v_2(3m_j + 1) = d_j$ holds automatically; the sole remaining condition is that the *final* formal iterate m_N be odd, which translates to the single additional bit condition $2^{S_N} m_N \equiv 2^{S_N} \pmod{2^{S_N+1}}$, i.e. the terminal-parity bit β_N selects between the two candidates χ_N and $\chi_N + 2^{S_N}$ inside the coarser class fixed by the aggregate congruence. Full details are given in [7, Prop. 5.2] and were re-verified independently, by exact-arithmetic brute force against direct orbit iteration, in the course of the present work. $\square$

Proposition 4.4 (Transport step and reset dichotomy; background, with a sharpened inequality). *Write $K_j := (T_j - 3^{-(j+1)}) \pmod{2^{d_{j+1}}} \in \{0, \dots, 2^{d_{j+1}} - 1\}$. Then*

$$\chi_{j+1} = \chi_j + 2^{S_j} K_j.$$

Consequently, writing $F_j := S_j - \log_2 \chi_j$,

$$K_j = 0 \implies F_{j+1} = F_j + d_{j+1}, \quad K_j \geq 1 \implies F_{j+1} < d_{j+1}, \quad (3)$$

the second inequality being strict.

Proof. The transport identity is [7, Prop. 5]; we reprove the strict form of the dichotomy, which is used repeatedly below. If $K_j = 0$ then $\chi_{j+1} = \chi_j$, giving $F_{j+1} = S_{j+1} - \log_2 \chi_j = F_j + d_{j+1}$ directly. If $K_j \geq 1$ then $\chi_{j+1} = \chi_j + 2^{S_j} K_j \geq 2^{S_j} + \chi_j > 2^{S_j}$, using $\chi_j \geq 1$ (Lemma 4.2); hence $\log_2 \chi_{j+1} > S_j$, so $F_{j+1} = S_{j+1} - \log_2 \chi_{j+1} < S_{j+1} - S_j = d_{j+1}$. $\square$

5 Pinning and launch-state collapse

5.1 The truncation lemma and terminal-suffix theorem

Lemma 5.1 (Truncation). *If $r = r(D)$ realizes D , then for every $1 \leq j \leq N$, $\chi_j = r \bmod 2^{S_j}$.*

Proof. Since r realizes every prefix D_j , it lies in the exact realizer class of D_j modulo 2^{S_j+1} (Proposition 4.3 applied at j); the coarse reduction of that class modulo 2^{S_j} is χ_j , and χ_j is by definition the unique representative of that reduction in $[0, 2^{S_j})$. $\square$

Throughout, write $p := \lfloor \log_2 \chi_N \rfloor$ (well defined and ≥ 0 by Lemma 4.2), and

$$t_1 := \min\{j \geq 1 : S_j > p\}, \quad \tau_N := t_1 - 1.$$

Lemma 5.2 (Pinning index). *$\chi_j = \chi_N$ for every $j \geq t_1$, and $K_{\tau_N} > 0$ whenever $\tau_N \geq 1$ (a genuine reset occurs immediately before t_1).*

Proof. For $j \geq t_1$, $S_j > p$ means $2^{S_j} > 2^p$, and since $p = \lfloor \log_2 \chi_N \rfloor$ we have $\chi_N < 2^{p+1} \leq 2^{S_j}$; so χ_N is already its own reduction modulo 2^{S_j} , giving $\chi_j = \chi_N \bmod 2^{S_j} = \chi_N$ (using r realizes D , Lemma 5.1, so $\chi_j = r \bmod 2^{S_j} = \chi_N \bmod 2^{S_j}$ whenever $S_j \leq S_N$, and equality with χ_N itself follows once $2^{S_j} > \chi_N$). If $\tau_N \geq 1$: by minimality of t_1 , $S_{\tau_N} \leq p < S_{t_1}$, so $2^{S_{\tau_N}} \leq 2^p \leq \chi_N$; since $\chi_{\tau_N} < 2^{S_{\tau_N}}$ strictly (Lemma 4.2), $\chi_{\tau_N} < 2^{S_{\tau_N}} \leq \chi_N = \chi_{t_1}$, so $\chi_{\tau_N} \neq \chi_{t_1}$, forcing $K_{\tau_N} \neq 0$ by Proposition 4.4. $\square$

Theorem 5.3 (Terminal-suffix depth identity). *Let $Q_N := S_N - S_{t_1}$. Then, exactly,*

$$F_N = F_{t_1} + Q_N, \quad \text{and if } \tau_N \geq 1: \quad 0 < F_{t_1} < d_{t_1}.$$

If $\tau_N = 0$ (no reset before $t_1 = 1$) then $0 < F_1 \leq d_1$, with equality only in the degenerate family $r(D) = 1$.

Proof. $F_N = S_N - \log_2 \chi_N = S_N - \log_2 \chi_{t_1}$ (Lemma 5.2) $= (S_N - S_{t_1}) + (S_{t_1} - \log_2 \chi_{t_1}) = Q_N + F_{t_1}$. For the bound on F_{t_1} : when $\tau_N \geq 1$, the proof of Lemma 5.2 shows $\chi_{t_1} = \chi_N > 2^{S_{\tau_N}} = 2^{S_{t_1} - d_{t_1}}$ (strict, since χ_N is odd by Lemma 4.2 and $2^{S_{\tau_N}}$ is a power of 2, so equality is impossible for $S_{\tau_N} \geq 1$; for $S_{\tau_N} = 0$ oddness of $\chi_N \geq 1$ still gives strictness), so $\log_2 \chi_{t_1} > S_{t_1} - d_{t_1}$, i.e. $F_{t_1} < d_{t_1}$. And $F_{t_1} > 0$ since $\chi_{t_1} < 2^{S_{t_1}}$ by definition. When $\tau_N = 0$, $\chi_1 < 2^{S_1} = 2^{d_1}$ gives $F_1 > 0$ directly, and $\chi_1 \geq 1$ gives $F_1 \leq d_1$, with equality iff $\chi_1 = 1$, which (tracing through $r(D) = \chi_N = \chi_1$ in this no-reset case) occurs exactly in the degenerate family $r(D) = 1$. $\square$

This identity — proved here directly from the truncation lemma — recovers, with sharp non-asymptotic constants, the reset-dichotomy consequence that large coarse depth F_N cannot accumulate through repeated small errors: it is, exactly, the valuation mass Q_N of a single uninterrupted zero-deficit (“perfect transport”) run at the end of the word, up to the bounded correction $F_{t_1} \in (0, d_{t_1})$.

5.2 Launch-state collapse

Theorem 5.4 (Launch-state collapse and forced-continuation equivalence). *Let j be any index with $2^{S_j} > r(D)$ (in particular $j \geq t_1$), and let*

$$m_j^* := \frac{3^j r(D) + C_j(D)}{2^{S_j}} \in \mathbb{Z}_{>0}$$

(an integer, being the formal j -th iterate under the aggregate congruence, and in fact the genuine j -th accelerated iterate of $r(D)$, by Proposition 4.3). Then

$$T_j = -m_j^* 3^{-j} \quad \text{in } \mathbb{Z}_2,$$

and the letters $(d_{j+1}, d_{j+2}, \dots, d_N)$ following index j are exactly the accelerated Collatz valuation word of m_j^ : $d_{j+i} = v_2(3\mu_{i-1} + 1)$ where $\mu_0 = m_j^*$, $\mu_i = \mathcal{C}(\mu_{i-1})$.*

Proof. By Lemma 5.1, $\chi_j = r \bmod 2^{S_j} = r$ (using $2^{S_j} > r$), so

$$T_j = \frac{\xi_j - \chi_j}{2^{S_j}} = \frac{-C_j 3^{-j} - r}{2^{S_j}} = -3^{-j} \cdot \frac{C_j + 3^j r}{2^{S_j}} = -3^{-j} m_j^*,$$

using the definition of ξ_j and the definition of m_j^* . That m_j^* is the genuine (not merely formal) accelerated iterate of $r(D)$ is immediate from Lemma 3.2, since $r(D)$ realizes D and $2^{S_j} m_j^* = 3^j r(D) + C_j(D)$ is exactly (1) at index j with $m_0 = r(D)$. Since $r(D)$'s own orbit realizes the entire word D by definition, its letters beyond index j are, by definition of the orbit, exactly the accelerated valuation word of its j -th iterate m_j^* . $\square$

Remark 5.5. *The forced-continuation equivalence is not an approximation or a heuristic reading of the transport recursion: it is a direct consequence of the fact that $r(D)$ is a genuine positive integer realizing D , together with the elementary algebraic identity relating T_j to the aggregate carry sum. No additional hypothesis (periodicity, confinement, or otherwise) is used.*

Definition 5.6 (Launch renormalization). *For a word D pinned at $t_1 = \tau_N + 1$, we call the passage*

$$D \mapsto (t_1, m_{t_1}^*, (d_{t_1+1}, \dots, d_N) = \text{orbit word of } m_{t_1}^*)$$

the launch renormalization of D at its terminal pinning index. This is purely an operational name for Theorem 5.4 applied at $j = t_1$; see §2 for the terminology audit.

6 Conditional obstruction architecture

Definition 6.1 (Dangerous word). *Fix a confinement width $c \geq 1$ and $\varepsilon \in (0, 1)$. A c -confined word D_N (i.e. $R_j := S_j - j \log_2 3 \leq c$ for all $0 \leq j \leq N$) with $r(D_N) \geq 3$ is ε -dangerous if*

$$\log_2 r(D_N) < \varepsilon N.$$

Remark 6.2 (Relation to the terminal-cylinder form, and an honesty caveat). *In Sector A ($\beta_N = 1$), Theorem 5.3 gives $\log_2 r = S_N - Q_N - F_{t_1}$ with $0 < F_{t_1} < d_{t_1}$, so ε -danger and the cylinder-depth condition $Q_N > S_N - \varepsilon N$ are equivalent up to the bounded reset window d_{t_1} . The two notions are not equivalent in Sector B: a word with $\beta_N = 0$ can have a very deep coarse terminal cylinder (small χ_N , large Q_N) while its actual least realizer $r = \chi_N + 2^{S_N}$ is enormous. Since the object of interest here is a floor on $r(D)$, the realizer-based Definition 6.1 is the correct one, and Sector-B words are irrelevant to it by the exact observation of Proposition 6.3(a) below.*

Proposition 6.3 (Necessary conditions for a dangerous word). *Suppose D_N is ε -dangerous. Then, writing $\alpha = \log_2 3$:*

- (a) $\beta_N = 1$, exactly and for every N : if $\beta_N = 0$ then $\log_2 r(D_N) \geq S_N \geq N > \varepsilon N$ (using $\varepsilon < 1$ and $d_i \geq 1$), contradicting Definition 6.1 directly. Hence $r(D_N) = \chi_N$.
- (b) $\tau_N \leq S_{\tau_N} \leq \lfloor \log_2 r(D_N) \rfloor < \varepsilon N$, hence $t_1 \leq \varepsilon N + 1$; moreover $S_{t_1} \leq \lfloor \log_2 r(D_N) \rfloor + d_{t_1} < \varepsilon N + d_{t_1}$, so $Q_N > S_N - \varepsilon N - d_{t_1}$.
- (c) $R_{t_1} \geq (1 - \alpha)t_1 > -(\alpha - 1)(\varepsilon N + 1)$.
- (d) **(Launch-size bound.)** If the orbit segment $r(D_N) = m_0, m_1, \dots, m_{t_1}$ is injective (which, by §8, fails only when the orbit meets a cycle, a case routed to the cycle branch), then

$$\log_2 m_{t_1}^* \leq \alpha \varepsilon N + \frac{1}{9} \log_2 N + O(1).$$

The leading coefficient is $\alpha \varepsilon$, not ε : the term $-R_{t_1} \leq (\alpha - 1)(\varepsilon N + 1)$ is linear in N and must not be absorbed into a logarithmic error.

- (e) The forced suffix, i.e. the orbit of $m_{t_1}^*$, has length $L := N - t_1 \geq (1 - \varepsilon)N - 1$ and remains, relative to its own starting drift, within a corridor of exact width

$$c' = c - R_{t_1} \leq c + (\alpha - 1)(\varepsilon N + 1).$$

Proof. (a) as stated; note that no largeness of N is required, only $\varepsilon < 1$. (b): $S_{\tau_N} \geq \tau_N$ since each letter is ≥ 1 ; $S_{\tau_N} \leq p = \lfloor \log_2 \chi_N \rfloor = \lfloor \log_2 r \rfloor$ by the definition of t_1 and (a); and $S_{t_1} = S_{\tau_N} + d_{t_1}$. (c) is the elementary bound $R_j \geq (1 - \alpha)j$ (each step changes R by $d_i - \alpha \geq 1 - \alpha$) combined with (b). (d): by the invariant of Lemma 6.4 below applied at $i = t_1$ with $m_0 = r(D_N)$,

$$\log_2 m_{t_1}^* = \log_2 r(D_N) - R_{t_1} + E_{t_1} < \varepsilon N + (\alpha - 1)(\varepsilon N + 1) + E_{t_1} = \alpha \varepsilon N + (\alpha - 1) + E_{t_1},$$

and $E_{t_1} \leq \frac{1}{9} \log_2 t_1 + O(1) \leq \frac{1}{9} \log_2 N + O(1)$ by the injective-orbit carry bound of [4, Lemma 4.4] (a self-contained weaker bound $E_{t_1} \leq \frac{1}{6} \log_2 t_1 + \frac{2}{3}$, sufficient here, follows from $E_i \leq \frac{1}{3 \ln 2} \sum_{k < i} 1/m_k$ and the harmonic sum over i distinct odd positive integers). (e) is confinement $R_j \leq c$ for $j \geq t_1$, rewritten in terms of the suffix's own drift $R_j - R_{t_1}$, combined with (c). $\square$

Lemma 6.4 (Orbit-floor invariant; known, Eliahou–Rozier). *For every odd m_0 and every $i \geq 0$ along its accelerated orbit,*

$$\log_2 m_i + R_i - E_i = \log_2 m_0, \quad E_i := \sum_{k < i} \log_2 \left(1 + \frac{1}{3m_k} \right) \geq 0.$$

Proof. This is the exponentiated form of $m_i 2^{R_i} = m_0 \prod_{k < i} (1 + \frac{1}{3m_k})$, which follows directly from $m_{k+1} = \frac{3m_k}{2^{d_{k+1}}} (1 + \frac{1}{3m_k})$ multiplied over $k < i$ and $3^i 2^{-S_i} = 2^{-R_i}$. This telescoped drift identity is standard in this literature: it is the log form of the affine identity (1) for $j = i$, m_0 fixed, and appears, in essentially this form, in the treatment of nontrivial-cycle bounds by Eliahou [8] (cyclic case) and in the general-trajectory reformulation of Rozier [12]; we do not claim novelty for it and use it as background. $\square$

Lemma 6.4 at $i = t_1$ with $m_0 = r(D)$, $m_{t_1} = m_{t_1}^*$ is what was used in the proof of Proposition 6.3(d): the launch integer is small, but on the scale $2^{\alpha \varepsilon N(1+o(1))}$, not $2^{\varepsilon N(1+o(1))}$ — the dive of the prefix (term $-R_{t_1}$) contributes a genuine linear factor $2^{(\alpha-1)\varepsilon N}$ in the worst case, attained by an all-ones prefix of length $\approx \varepsilon N$.

Definition 6.5 (Case split, at scale q_0). Fix a period cap $q_0 \in \mathbb{Z}_{\geq 1}$. The forced suffix of an ε -dangerous word — equivalently, the orbit of $m_{t_1}^*$ — falls into exactly one of:

- periodic tail (at scale q_0): the terminal letters form an exact repetition X^k , $k \geq 3$, of a block X of period $|X| \leq q_0$ that is not entirely 2's;
- cycle tail: the terminal letters ride the trivial fixed cycle (a run of the letter 2) or, hypothetically, a nontrivial accelerated cycle;
- irregular (at scale q_0): neither of the above.

The cap q_0 is necessary because the periodic obstruction of §7 carries block-dependent constants B_X , $n_0(X)$ that are bounded over $|X| \leq q_0$ for each fixed q_0 but are not bounded over blocks of unbounded length (in the worst case B_X can grow exponentially in $|X|$); repetitive tails of period growing with N therefore remain, honestly, in the irregular class.

7 The periodic-anchor obstruction

Proposition 7.1 (Anchor-cycle dichotomy). For a block $X = (x_1, \dots, x_q)$, $q \geq 1$, write $S_X = \sum x_i$ and $\delta_X := 3^q - 2^{S_X}$. Since $\log_2 3$ is irrational, $\delta_X \neq 0$ for every $q \geq 1$; since 3^q is odd and 2^{S_X} is even, δ_X is odd, hence a unit in $\mathbb{Z}_2$, and $\xi_X := -C(X)/\delta_X \in \mathbb{Q} \cap \mathbb{Z}_2$ is well defined. Then ξ_X is a positive integer if and only if ξ_X is the minimum element of a genuine positive accelerated Collatz cycle with valuation word X ; and in that case ξ_X is the unique such minimum.

Proof. This is the accelerated-word specialization of the correspondence of Böhm and Sontacchi [3], who established the analogous fact for the (un-accelerated) binary parity vector: a candidate cyclic admissibility pattern determines a unique rational solution to the cyclic form of the affine identity, and that solution is a genuine cycle exactly when it is a positive integer. In the present notation, the cyclic case of Lemma 3.2 ($m_0 = m_q =: m$) reads $2^{S_X} m = 3^q m + C(X)$, i.e. $m\delta_X = -C(X)$, i.e. $m = \xi_X$; since $C(X) = \sum_{i=0}^{q-1} 3^{q-1-i} 2^{s_i} > 0$ is manifestly a sum of positive terms, a solution $m > 0$ requires $\delta_X < 0$, i.e. $2^{S_X} > 3^q$, i.e. genuine positive drift $S_X > q \log_2 3$ over the block. Conversely, if $\xi_X \in \mathbb{Z}_{>0}$, the same integrality-and-terminal-parity argument used in the proof of Proposition 4.3 (applied to the cyclic aggregate congruence, all indices modulo q) shows every formal iterate of ξ_X under one application of X is a positive odd integer with exactly the prescribed valuations, so ξ_X 's orbit is periodic with word X ; uniqueness follows since ξ_X is by construction the unique solution of the linear equation $m\delta_X = -C(X)$. $\square$

Remark 7.2. Proposition 7.1 is exactly the excluded case of the periodic realizer floor established in the author's companion note [5]: that note proves, for $\xi_X \notin \mathbb{Z}_{>0}$, a floor $r(X^n) \geq 2^{nS_X - B_X}$ for an explicit finite B_X (the eventual binary period length of ξ_X 's expansion), by a fixed rational 2-adic anchor mechanism. Proposition 7.1 identifies precisely why the hypothesis $\xi_X \notin \mathbb{Z}_{>0}$ cannot be dropped: exactly the excluded blocks are those realizing genuine cycles, for which $r(X^n) = \xi_X$ stays bounded as $n \rightarrow \infty$ and the floor of [5] fails by construction (not merely by an artifact of the proof method).

Theorem 7.3 (Periodic-tail obstruction). Suppose D is c -confined, $r(D)$ realizes an injective orbit segment, and the last $q \cdot n$ letters of D (starting at index $i = N - qn$) form X^n for a block X with $\xi_X \notin \mathbb{Z}_{>0}$. Then

$$\log_2 r(D) \geq (S_N - S_i) - (\alpha - 1)i - B_X - \frac{1}{9} \log_2 N - O(1). \quad (4)$$

Consequently, an ε -dangerous word's exactly periodic non-cycle tail can begin no earlier than

$$i \geq \frac{1-\varepsilon}{\alpha} N - O_X(\log N),$$

i.e. it can cover at most a fraction

$$\frac{N-i}{N} \leq 1 - \frac{1-\varepsilon}{\alpha} + O_X\left(\frac{\log N}{N}\right) = 1 - (1-\varepsilon) \log_3 2 + O_X\left(\frac{\log N}{N}\right)$$

of the word. In the frontier limit $\varepsilon \rightarrow 0$ this fraction tends to $1 - \log_3 2 \approx 0.36907$; for finite ε the bound is strictly weaker, by exactly ε/α .

Proof. Let m_i be the (genuine) i -th accelerated iterate of $r(D)$; since D 's suffix from index i realizes X^n , $m_i \geq r(X^n) \geq 2^{nS_X - B_X} = 2^{(S_N - S_i) - B_X}$ by the periodic realizer floor of [5]. By the orbit-floor invariant (Lemma 6.4), $\log_2 r(D) = \log_2 m_i + R_i - E_i$, and $R_i \geq (1-\alpha)i$, $E_i \leq \frac{1}{9} \log_2 N + O(1)$ (injective-orbit carry bound, [4, Lemma 4.4]). Combining,

$$\log_2 r(D) \geq [(S_N - S_i) - B_X] + (1-\alpha)i - \frac{1}{9} \log_2 N - O(1),$$

which is (4). For the fraction bound: danger gives $\log_2 r(D) < \varepsilon N$, and $S_N - S_i \geq N - i$ (each $d_j \geq 1$); substituting into (4): $\varepsilon N > (N-i) - (\alpha-1)i - B_X - O(\log N) = N - \alpha i - B_X - O(\log N)$, hence $\alpha i > (1-\varepsilon)N - B_X - O(\log N)$, i.e. $i \geq \frac{1-\varepsilon}{\alpha} N - O_X(\log N)$, and the tail-length statement follows by complementation. The block-dependent constants B_X , $n_0(X)$ are absorbed into $O_X(\cdot)$; the bound is therefore uniform over any fixed finite family of blocks (in particular over $|X| \leq q_0$ for fixed q_0), but not over blocks of unbounded length, as recorded in Definition 6.5. $\square$

Remark 7.4 (On the constant). We record the numerical coincidence that $1 - \log_3 2$ is exactly the complement of $\log_3 2$, the exponent appearing in the entropy-scale calibration of the residue-axis problem in [4]; the derivation above is purely a drift-arithmetic consequence of Proposition 4.4's dichotomy and the wedge bound, with no counting or entropy argument entering at any step. We do not interpret this coincidence beyond what is proved: it may reflect a genuine connection between the drift and residue axes of the underlying occupation problem, or it may be an artifact of both quantities being governed by $\alpha = \log_2 3$ for unrelated reasons. No further claim is made.

8 The cycle/drift obstruction

Lemma 8.1 (Cycle drift positivity; known). *Every nontrivial accelerated Collatz cycle has strictly positive drift per period: if the cycle has word X of length q and total valuation S_X , then $S_X > q \log_2 3$.*

Proof. This is the content of Proposition 7.1's proof, specialized to $\xi_X \in \mathbb{Z}_{>0}$: a genuine cycle forces $\delta_X = 3^q - 2^{S_X} < 0$. This mechanism — the cyclic affine identity forcing $2^{S_X} > 3^q$ — is standard in the cycle-bound literature following Böhm and Sontacchi [3] and is used explicitly by Eliahou [8] to derive Diophantine-approximation lower bounds on nontrivial cycle length. $\square$

Theorem 8.2 (Trivial-cycle branch: empty or bounded). *Let D_N be c -confined and realized by $m_0 \geq 3$, and suppose the orbit of m_0 reaches 1 at some index $h \leq N$ (so that the letters after h ride the trivial cycle). Then*

$$R_h = \log_2 m_0 + E_h \geq \log_2 m_0.$$

Consequently: (i) if $c < \log_2 m_0$, this is incompatible with confinement, so for $m_0 = r(D) \geq 3$ and any $c < \log_2 3 \approx 1.585$ the trivial-cycle branch is empty; (ii) for general c , the branch requires $r(D) \leq 2^c$, a finite family of realizers, each with a finite maximal c -confined orbit length, so the branch contributes words of length at most a constant N_c depending only on c . In either case, once $N > N_c$, no ε -dangerous word has a trivial-cycle tail, and the drift bound $t \leq (c - R_h)/(2 - \alpha)$ (each letter 2 raising R by exactly $2 - \alpha > 0$) bounds any residual riding within the finitely many exceptional words.

Proof. The identity $R_h = \log_2 m_0 - \log_2 m_h + E_h$ is Lemma 6.4; $m_h = 1$ and $E_h \geq 0$ give $R_h \geq \log_2 m_0$. Confinement demands $R_h \leq c$, forcing $\log_2 m_0 \leq c$, i.e. $m_0 \leq 2^c$: finitely many odd realizers. Each such m_0 has a finite maximal c -confined prefix length (its drift eventually exceeds c , since its orbit reaches 1 and thereafter R increases by $2 - \alpha$ per step without bound), giving $N_c < \infty$. This was additionally verified by exact computation at the first hit of 1 for every odd seed below 2×10^4 (9999 cases, no violations). $\square$

Remark 8.3 (Unknown nontrivial cycles: what is, and is not, excluded). *An earlier draft of this note claimed that existing cycle exclusions render a hypothetical nontrivial-cycle tail “incompatible with danger at any fixed ε once N is large.” That claim was incorrect, with the asymptotics running in the wrong direction, and is retracted. The three available tools prove the following, and no more:*

1. Positive drift per period (Lemma 8.1) constrains a tail only once it completes full periods. By Hercher [10], every nontrivial cycle has at least 92 local minima, and the odd-step period exceeds 7.2×10^{10} by the bounds collected there; any cycle-riding tail of length below that period is a partial arc, along which the drift can locally decrease, so the per-period mechanism gives no bound at any presently relevant word length.
2. Computational minimum-member bounds [1] give $m_{\min} > 2^{68}$ for any nontrivial cycle. If a dangerous word’s suffix meets such a cycle at index i , then $m_i \geq m_{\min}$ and Lemma 6.4 gives $\log_2 r(D) \geq 68 + R_i - E_i \geq 68 - (\alpha - 1)N - O(\log N)$. This contradicts $\log_2 r(D) < \varepsilon N$ only for $N \lesssim 68/((\alpha - 1) + \varepsilon)$; for larger N the size argument is vacuous, because a long confined dive can absorb the required magnitude.
3. Cycle-structure exclusions [13, 10] bound which cycles can exist but say nothing further about partial arcs of an unexcluded one.

The honest position is therefore: the cycle branch is closed unconditionally (and sharply, by Theorem 8.2) for the trivial cycle, and open for hypothetical nontrivial cycles at large N . Rather than assume nonexistence silently, the conditional theorem of §11 carries the hypothesis explicitly; equivalently, any ε -dangerous word evading it through this branch would itself certify a nontrivial accelerated cycle — an event that the cycle literature treats as a separate, independently studied phenomenon.

9 Launch renormalization and the irregular residual class

Proposition 9.1 (Renormalization/Residual Proposition). *Let D_N be ε -dangerous, $N > N_c$, and assume no nontrivial accelerated cycle meets the orbit segment of $r(D_N)$ (Remark 8.3). Fix a period cap $q_0 \geq 1$ below the current cycle-period bounds collected in [10]. By Proposition 6.3, the forced suffix W (from t_1 to N) is exactly the accelerated orbit of the launch integer $m_{t_1}^*$ (Theorem 5.4), with $\log_2 m_{t_1}^* \leq \alpha \varepsilon N + O(\log N)$ and length $L \geq (1 - \varepsilon)N - 1$, remaining within a corridor of exact*

width $c' = c - R_{t_1} \leq c + (\alpha - 1)(\varepsilon N + 1)$ relative to its own start, and injective. Moreover, every exact terminal repetition X^k of W with $|X| \leq q_0$, $k \geq 3$, covers at most a fraction

$$\theta^*(\varepsilon) + O_{q_0}\left(\frac{\log N}{N}\right), \quad \theta^*(\varepsilon) := \frac{\alpha - 1 + \varepsilon}{\alpha(1 - \varepsilon)},$$

of the suffix length L ; in particular this fraction is below $\frac{1}{2}$ whenever $\varepsilon < \frac{2-\alpha}{2+\alpha} \approx 0.11577$ and N is large.

Proof. The launch data are Proposition 6.3(d)–(e) and Theorem 5.4; injectivity holds because a repeated state would place the segment on a cycle, excluded by Theorem 8.2 (trivial, $N > N_c$) and by hypothesis (nontrivial). For the fraction bound, let the terminal repetition be X^k , of length ℓ , with $|X| \leq q_0$. Three cases. (1) $\xi_X \notin \mathbb{Z}_{>0}$: Theorem 7.3 gives $\ell \leq N^{\frac{\alpha-1+\varepsilon}{\alpha}} + O_X(\log N)$; dividing by $L \geq (1 - \varepsilon)N - 1$ gives the stated fraction. (2) X a power of the letter 2 (trivial-cycle block, $\xi_X = 1$): the realizers of $(2)^t$ form the class of 1 modulo 2^{2t+1} , so the iterate at the repetition's start is 1 (excluded, $N > N_c$) or $\geq 1 + 2^{2t+1}$; Lemma 6.4 then gives $\varepsilon N > \log_2 r \geq 2t - (\alpha - 1)N + \dots$, i.e. $t \leq N^{\frac{\alpha-1+\varepsilon}{\alpha+1}} + O(\log N)$, a smaller fraction still. (3) $\xi_X \in \mathbb{Z}_{>0}$ with X not a power of (2): by Proposition 7.1, X would be the word of a genuine nontrivial cycle of period $\leq q_0$, excluded unconditionally by the cycle-length bounds of [13, 10] for any q_0 below their range. Finally, $\theta^*(\varepsilon) < \frac{1}{2} \iff 2(\alpha - 1 + \varepsilon) < \alpha(1 - \varepsilon) \iff \varepsilon < \frac{2-\alpha}{2+\alpha}$. $\square$

Remark 9.2 (Why no “irregular core” is defined). *An earlier version of this note removed the maximal terminal repetition and asserted that the remainder was irregular. That step was flawed: a suffix of the schematic form (irregular) $Y^3 X^4$ loses only X^4 under a single removal and still ends in the repetition Y^3 , and iterating the removal is not controlled by the present estimates — applying Theorem 7.3 to successive prefixes (valid, since r of a prefix is at most r of the word) yields peel boundaries obeying $i_{k+1} \geq i_k/\alpha - \varepsilon N/\alpha - O(\log N)$, a recursion whose guaranteed floor decays geometrically and becomes vacuous after $\approx \log_\alpha(1/\varepsilon)$ layers, so nested repetition towers covering almost the entire suffix are not excluded. We therefore make no claim about an irregular residual core; the supportable statement is the fractional one of Proposition 9.1, which inspects only the outermost terminal repetition and is exactly what the conditional theorem of §11 consumes.*

Remark 9.3 (The induced parameter flow: a proved two-sided envelope). *The reduction is a genuine self-map of the problem class: once the forced suffix is long enough, $m_{t_1}^*$ is the least realizer of the suffix word (it lies in the suffix's realizer class and is below the relevant modulus), so the renormalized datum (suffix word, $m_{t_1}^*$) has the same form as $(D_N, r(D_N))$. In the dimensionless coordinate $\rho = \log_2 r/N$ (danger parameter), one renormalization step satisfies the proved two-sided envelope*

$$\rho - \frac{c}{N} \leq \rho' \leq \frac{\alpha\rho}{1-\rho} + o(1),$$

where the lower bound is the confined-orbit floor: $R_{t_1} \leq c$ and Lemma 6.4 give $m_{t_1}^* \geq r \cdot 2^{-R_{t_1}} \geq r \cdot 2^{-c}$, so $\log_2 m_{t_1}^* \geq \rho N - c$, and dividing by the new length $L \leq N$ preserves the bound. We are careful about what this does and does not establish. It does establish non-contraction: the danger parameter cannot decrease by more than $c/N = o(1)$ per step, so the frontier $\rho = 0$ cannot be approached under iteration, and no contraction-type obstruction is available from these estimates. It does not establish expansion: the upper bound is only an envelope, and whether ρ' actually grows (by the factor $\alpha/(1 - \rho)$ the envelope permits, or at all) is not determined by the inequalities proved here. Likewise, after k steps the certified range of the parameters is governed by iterating the envelope, which exceeds the applicability threshold $\varepsilon < \frac{2-\alpha}{2+\alpha}$ of Proposition 9.1 within roughly

$\log_\alpha(1/\varepsilon)$ iterations — meaning only that the present estimates cease to certify the reduction beyond that depth, not that the reduction itself degenerates. Iterated launch renormalization, as controlled by these bounds, is therefore a finitely certifiable, non-contracting reformulation: self-similarity of form is proved, together with the impossibility of parameter contraction; genuine parameter expansion remains an open (and, for the obstruction program, moot) question.

This is the precise sense in which the residual irregular class is *renormalization-stable*: it is not merely unclassified, but demonstrably contains, for every N , a rescaled instance of the same kind of object (a small integer with a long, near-critically-confined injective orbit segment) that generated the original problem. In particular:

Corollary 9.4. *Finite structural enumeration alone — classifying dangerous words by symbolic pattern (periodic, eventually periodic, bounded defects from a periodic block, cycle-riding, or any finite union of such classes) and treating each class with a class-specific obstruction — cannot, without an independent obstruction valid on the residual irregular class, close the conditional-obstruction program: the forced suffix of a dangerous word in that residual class is itself an object of the original kind (a small integer with a long, near-critically confined injective orbit segment), so each round of classification reproduces an instance of the class it has not obstructed. This does not preclude that some finite classification might one day be paired with a theorem controlling all irregular members of one of its classes; it asserts only that the enumeration step itself supplies no such theorem, and that the burden it leaves is exactly the hypothesis of Definition 11.2.*

Remark 9.5. *We emphasize, per the discussion in §10 below, that Proposition 9.1 does not assert that the residual problem is easier, harder, or the same size as the original; it asserts only self-similarity of form. §11 shows that, once the corridor scaling is accounted for correctly, the residual problem is in fact at least as demanding as the fixed-corridor original.*

10 Minimum informative representations and provenance

The launch-state analysis above sits naturally inside the target-relative representation-sufficiency framework developed in the author’s companion note [6], which formalizes when a reduced representation π of a state x retains enough information for a target $Q(x)$, via the target fiber diameter $\Delta_Q(y) = \sup\{|Q(x) - Q(x')| : \pi(x) = \pi(x') = y\}$. Applying that audit to the present setting, with target $Q(D) = \log_2 r(D) - I_{\text{Collatz}}N$:

- The coarsest representation, (N, S_N) , has enormous fiber diameter in Q : essentially the full entropy-scale range of $\log_2 r(D)$ over all confined words at that (N, S_N) .
- Adding β_N (one bit) roughly halves the diameter, by Proposition 6.3(a): it separates Sector A from the frontier-irrelevant Sector B.
- Adding Q_N reduces the fiber diameter of Q to essentially the bounded reset window $F_{t_1} \in (0, d_{t_1})$, by Theorem 5.3 and Proposition 4.3. This achieves near-exact sufficiency — but Q_N is, by the truncation lemma (Lemma 5.1), a deterministic function of the binary expansion of $r(D)$ itself on the valuation grid $\{S_j\}$; conditioning on it and observing that the residual variation is small is therefore *circular* as an explanation of why $r(D)$ is small, in the technical sense of [6]’s provenance gate: the representation’s sufficiency for the target is not independent evidence, because the representation is (up to a bounded correction) a re-encoding of the target.

- Adding the full launch integer $m_{t_1}^*$ achieves *exact* sufficiency (Theorem 5.4) — but at the cost of reproducing the original dynamical object in full: it is an exact, non-compressive reparameterization, not a reduction. This is the representation-theoretic restatement of Corollary 9.4.

The remaining methodological objective, precisely stated, is therefore: identify an independently computable coarse invariant of the launch state — weaker than the full integer $m_{t_1}^*$, and not reducible to Q_N or $r(D)$ by the truncation lemma — that is nonetheless strong enough to obstruct long irregular near-critical confinement. Theorems 7.3 and 8.2 are exactly two instances where such an invariant (the block period X , or the letter-2 run length) was found; §11 states precisely what would be needed for the irregular class.

11 Conditional implication for an exponential realizer floor

The natural closing hypothesis proposed in earlier stages of this program was stated informally as: *no positive integer has an injective accelerated orbit segment remaining within a fixed near-critical corridor for more than $K \log_2 m / \varepsilon$ steps*. We now audit this statement’s quantifiers against Proposition 6.3(e), and find that the corridor cannot be taken fixed.

Proposition 11.1 (Corridor-scaling correction). *The corridor width available to the forced suffix in Proposition 9.1 is exactly $c' = c - R_{t_1}$, bounded by*

$$c' \leq c + (\alpha - 1)(\varepsilon N + 1),$$

which is $\Theta(\varepsilon N)$, not $O(1)$, as $N \rightarrow \infty$ at fixed ε . Consequently a hypothesis of the form “no orbit sustains injective confinement within a fixed corridor for more than $f(m)$ steps” is, taken literally, insufficient to close the residual branch: what is required is uniformity in the corridor width A , at (at most) linear cost in A . Three regimes must be kept apart: (i) the original problem has a fixed corridor $c = O(1)$; (ii) the renormalized problem inherits an absolute corridor growing linearly, $c' \leq c + (\alpha - 1)\varepsilon N + O(1)$; (iii) the relative corridor, per unit of suffix length, remains small: $c'/L \leq (\alpha - 1)\varepsilon/(1 - \varepsilon) + o(1) = O(\varepsilon)$. Launch renormalization therefore preserves the form of the confinement problem while changing its normalization span: the daughter problem is near-critical in relative terms but is posed over a corridor that is unbounded in absolute terms.

Proof. Immediate from Proposition 6.3(c),(e): $c' = c - R_{t_1}$ exactly, $R_{t_1} > -(\alpha - 1)(\varepsilon N + 1)$, and $L \geq (1 - \varepsilon)N - 1$ for the ratio. $\square$

Definition 11.2 (Corridor-uniform lifetime hypotheses). *For $K < \infty$, consider the following open conditions on the accelerated dynamics.*

- $U_{\text{all}}(K)$: *for every odd $m \geq 3$ and every $A \geq 1$, every injective initial orbit segment of m satisfying $R_i(m) \leq A$ for $0 \leq i \leq L$ has $L \leq K(\log_2 m + A)$.*
- $U_{\text{half}}(K, q_0)$: *the same bound, assumed only for segments whose valuation word’s exact terminal repetitions of period $\leq q_0$ (with at least 3 repeats) each cover at most half the segment length.*

*Clearly $U_{\text{all}}(K) \Rightarrow U_{\text{half}}(K, q_0)$ for every q_0 : the latter exempts from the lifetime bound every orbit segment whose word is majority-covered by a single bounded-period terminal repetition. Both are **open conditional hypotheses**, not results of this note; nothing here is evidence for either. (An earlier draft used a hypothesis class defined by the absence of any terminal repetition; that*

formulation fails under nested repetitions, as recorded in Remark 9.2, and is replaced by the fractional condition here, which inspects only the outermost terminal repetition and is exactly what Proposition 9.1 certifies for dangerous words.)

Proposition 11.3 (Uniform lifetime bound $\Rightarrow$ exponential realizer floor, with explicit thresholds). *Fix $c \geq 1$ and assume no nontrivial accelerated cycle exists (Remark 8.3; alternatively, restrict all statements to words whose realizer orbits avoid such cycles).*

(i) *If $U_{\text{all}}(K)$ holds, then for every*

$$\varepsilon < \varepsilon_0(K) := \frac{1}{1 + (2\alpha - 1)K}$$

there is $N_1 = N_1(c, \varepsilon, K)$ such that no c -confined word of length $N \geq N_1$ is ε -dangerous; equivalently $\log_2 r(D_N) \geq \varepsilon N$ for all such words with $r \geq 3$: an exponential least-realizer floor.

(ii) *If only $U_{\text{half}}(K, q_0)$ holds (for some fixed $q_0 \geq 1$ below the cycle-period bounds of [10]), the same conclusion holds for every*

$$\varepsilon < \min \left\{ \varepsilon_0(K), \frac{2-\alpha}{2+\alpha} \right\},$$

with N_1 now depending also on q_0 . Since $\varepsilon_0(K) < \frac{2-\alpha}{2+\alpha} \approx 0.11577$ already for $K > 3.53$, the second constraint is inactive at every realistic K , and the governing inequality is in both versions

$$1 - \varepsilon > K(2\alpha - 1)\varepsilon.$$

Numerically, $2\alpha - 1 = 2.1699\dots$; at the heuristically expected occupation-scale constant $K \approx 12.61$ one gets $\varepsilon_0 \approx 0.0353$ in both versions. The conclusion is nontrivial for every finite K : the admissible range of ε is nonempty, though it shrinks like $1/K$.

Proof. Suppose D_N is ε -dangerous, N large. (i): the entire forced suffix is the orbit of $m_{t_1}^*$, injective (trivial-cycle entry is excluded for $N > N_c$ by Theorem 8.2, nontrivial by hypothesis), of length $L \geq (1 - \varepsilon)N - 1$, within a corridor of width $A = c' \leq c + (\alpha - 1)(\varepsilon N + 1)$ measured from its own start (Proposition 11.1). Applying $U_{\text{all}}(K)$ with Proposition 6.3(d):

$$(1 - \varepsilon)N - 1 \leq L \leq K(\log_2 m_{t_1}^* + c') \leq K(\alpha \varepsilon N + (\alpha - 1)\varepsilon N) + O(K \log N) = K(2\alpha - 1)\varepsilon N + O(K \log N),$$

so $1 - \varepsilon \leq K(2\alpha - 1)\varepsilon + o(1)$, contradicting $\varepsilon < \varepsilon_0(K)$ for N large. (ii): by Proposition 9.1, since $\varepsilon < \frac{2-\alpha}{2+\alpha}$, every exact terminal repetition of period $\leq q_0$ in the forced suffix covers at most $\theta^*(\varepsilon)L + O_{q_0}(\log N) < \frac{1}{2}L$ for N large; the forced suffix therefore lies in the class on which $U_{\text{half}}(K, q_0)$ asserts the lifetime bound, and the computation of case (i) applies verbatim to the whole suffix, giving the same threshold $\varepsilon_0(K)$. No removal, peeling, or “irregular core” is used anywhere (cf. Remark 9.2). The floor statement is the contrapositive of Definition 6.1. $\square$

Remark 11.4 (What the periodic obstruction buys, and noncircularity). *The role of Theorem 7.3 (with the all-2 and integral-anchor cases handled in Proposition 9.1’s proof) is now exactly delimited: it is not needed for version (i); its entire contribution is to prove that dangerous suffixes lie in the fractional class of U_{half} , thereby weakening the required hypothesis from U_{all} to U_{half} — a genuinely smaller class of orbits, exempting every segment majority-covered by a bounded-period terminal repetition — at no cost in the threshold $\varepsilon_0(K)$ for $K > 3.53$. On noncircularity: the hypotheses*

U quantify over the orbits of all positive integers and are independent of any particular word D ; but restricted to the fixed corridor $A = c$, a lifetime bound of this type is equivalent, via the record-chronology inversion established in [4], to the least-realizer floor itself. The non-tautological content of Proposition 11.3 therefore lies precisely in its two refinements: the required uniformity in the corridor width A (the cost of the launch route, Proposition 11.1), and the exemption of majority-periodic-tailed orbits (the payoff of the periodic branch). We state this plainly so that the proposition is not misread as reducing the problem to something of a different, easier species.

The hypotheses U_{all} , U_{half} are labeled explicitly as **open conditional hypotheses**, not results of this note: they are the precise, quantifier-audited forms of the residual obstruction that the launch-renormalization program reduces the least-realizer floor to, and we claim no progress toward proving either. We record that, unlike informally stated versions from earlier stages of this program, the hypothesis must be uniform in the corridor width A , with the relevant instances having $A = \Theta(\varepsilon N)$ — making it, if anything, a marginally *harder* statement than the fixed-corridor single-window form of the original occupation problem, not an easier one; and that the constants have been made numerically explicit precisely so that any future partial result toward U can be checked against the threshold $\varepsilon_0(K)$ for nontriviality.

12 Discussion and open problem

We summarize the status of every result invoked in this note.

Proved here (new): the launch-state collapse and forced-continuation equivalence (Theorem 5.4); the terminal-suffix depth identity with sharp constants (Theorem 5.3); the accelerated-word anchor-cycle dichotomy (Proposition 7.1, adapting [3]); the ε -dependent periodic-tail obstruction, with frontier limit $1 - \log_3 2$ and block-uniformity scope made explicit (Theorem 7.3); the sharpened trivial-cycle result — that branch is empty for $r \geq 3$ once $c < \log_2 3$, and bounded in general (Theorem 8.2); the necessary-condition package with the corrected launch-size exponent $\alpha\varepsilon$ (Proposition 6.3); the Renormalization/Residual Proposition with the terminal-repetition fraction bound $\theta^*(\varepsilon) = \frac{\alpha-1+\varepsilon}{\alpha(1-\varepsilon)}$ and the proved two-sided (non-contracting, expansion-open) parameter envelope (Proposition 9.1, Remark 9.3, Corollary 9.4); the corridor-scaling correction with the three-regime analysis (Proposition 11.1); and the two-version conditional sufficiency implication, both versions with the explicit threshold $\varepsilon_0(K) = (1 + (2\alpha - 1)K)^{-1}$ (Proposition 11.3).

Established background, restated for self-containedness: the exact realizer congruence (Proposition 4.3) and the transport-step dichotomy (Proposition 4.4), both from [7]; the periodic realizer floor mechanism ([5]), used but not reproved in §7; the affine transport identity, the orbit-floor invariant, and the injective-orbit carry bound (Lemmas 3.2, 6.4, and the cited Lemma 4.4), all traceable to the Terras/Everett/Eliahou/Rozier line of work [15, 9, 8, 12] as detailed in §2, and further developed in [4].

Known results imported directly: the Böhm–Sontacchi cycle correspondence [3]; cycle-length exclusion ranges [13, 10]; the computational verification range [1].

Explicitly not established, or investigated and found unavailable: any obstruction from defect count, density, or cumulative discrepancy of a confined word’s departure from the trivial-cycle word (a direct counterexample — the record confinement word for the seed 27, with 29 non-2 letters among 39 and cumulative discrepancy -17 , yet near-maximal terminal depth — rules this out); any small-odd-modulus congruence obstruction interacting with the forced 2-adic structure (full residue occupancy was observed at every tested modulus along the largest available record orbits, a null result, not a proof of impossibility); any leverage from ordinary prime-support

(radical/ABC-style) information on the realizer, which prior work in this program found essentially independent of the relevant 2-adic structure.

Open: the hypotheses $U_{\text{all}}(K)$ and $U_{\text{half}}(K, q_0)$ of Definition 11.2, i.e. corridor-uniform lifetime bounds for injective accelerated orbits, remain unproved; and the nontrivial-cycle branch of §8 is open at large N (Remark 8.3), entering the final theorem as an explicit hypothesis rather than being excluded by current cycle bounds. Corollary 9.4 shows the residual gap is not one that finite structural enumeration alone can close, and Remark 9.3 shows that iterating the reduction provably cannot contract its governing parameter, while whether it genuinely expands remains undetermined.

13 Conclusion

We have given a rigorous decomposition of the accelerated-Collatz least-realizer/confinement problem: dangerous confined words admit a short free prefix and a forced suffix that is exactly the accelerated orbit of a small integer (Theorem 5.4), and this forced suffix splits into a bounded-period periodic component controlled by a fixed rational 2-adic anchor and its exceptional cycle case (§7, via the classical Böhm–Sontacchi correspondence), a cycle-riding component that is closed sharply for the trivial cycle and remains conditional for hypothetical nontrivial cycles (§8), and an irregular residual component that is provably renormalization-stable: it regenerates, at a smaller but comparably confined scale, the same kind of object that generated the original problem (Proposition 9.1). Once the corridor-scaling correction of §11 is accounted for, the residual problem is not a strictly smaller instance of the original one; it is, if anything, marginally more demanding.

No proof of the Collatz conjecture is claimed. The value of this note, if any, lies in the exact boundary it draws between the conditional obstructions that genuinely work — with explicit constants, and honest attribution to the classical literature that they build on — and the irregular residual problem that remains open, stated in a form whose quantifier dependence on the confinement width, the danger parameter, and the drift constant $\alpha = \log_2 3$ has been made explicit rather than left informal.

A Computational verification

Every identity and inequality with a discrete or finite-arithmetic character was checked by exact-arithmetic computation (Python arbitrary-precision integers; no floating-point step in any discrete check) prior to being stated as a theorem above, following the epistemic discipline of the companion technical notes [7, 4]: computational checks validate that a stated formula is correctly transcribed and free of sign or indexing errors, and are not substituted for the proofs given in the body of the note.

Acknowledgments and AI-Assisted Research Statement

This work was developed through an integrated collaboration between the author and large language models (LLMs), used as interactive research tools for symbolic derivation, computational verification, literature search, and manuscript preparation. The author supplied the research direction, the falsification criteria applied at each stage, and the final judgments recorded above about what is proved, what is background, and what remains open. Every theorem stated above was independently re-derived rather than accepted from an earlier round of this program on trust, and one claimed result from an earlier stage of the program (an alleged sparse-defect exponent-one

Checked identity	Trials	Result
Exact realizer congruence (Prop. 4.3) vs. brute-force orbit search	250	0 mismatches
Truncation lemma $\chi_j = r \bmod 2^{S_j}$ (Lemma 5.1)	250, all j	0 mismatches
Transport step / bit-window identity (Prop. 4.4)	250, all j	0 mismatches
Reset dichotomy, strict form (eq. (3))	250, all j	0 mismatches
Pinning-index characterization (Lemma 5.2)	250	0 mismatches
Terminal-suffix identity and bound (Thm. 5.3)	250	0 mismatches
Launch-state collapse $T_j = -m_j^* 3^{-j}$ (Thm. 5.4)	148, all pinned j	0 mismatches
Forced continuation = orbit of m_j^* (Thm. 5.4)	148	0 mismatches
Anchor-cycle dichotomy, exhaustive over blocks $q \leq 6$	all blocks	confirmed
Trivial-cycle tail bound (Thm. 8.2)	300 constructions	0 violations
Periodic-tail assembled inequality (eq. (4))	193 words	0 violations
Frontier records, exact recomputation, $c = 1$, seeds to 7×10^6	21 records	all matched published chr

Table 1: Summary of computational verification performed for this note. All checks used exact integer arithmetic.

recovery, and, separately, an alleged counterexample to the exact-realizer formula) was found in the course of this program to rest on an error and was retracted; neither appears in this note. A final adversarial audit of the quantitative chain prior to circulation additionally corrected four items in an earlier draft of this note itself: the launch-size exponent ($\varepsilon \rightarrow \alpha\varepsilon$), the periodic-tail fraction (frontier constant promoted to an ε -dependent bound, with the block-uniformity scope added), the residual-length coefficient ($\frac{1}{\alpha} - \varepsilon \rightarrow \frac{1}{\alpha} - \frac{\alpha+1}{\alpha}\varepsilon$), and the cycle branch (an incorrectly claimed asymptotic exclusion of hypothetical nontrivial cycles was retracted and replaced by the sharper trivial-cycle result together with an explicit hypothesis). These corrections are recorded here rather than silently absorbed. Bibliographic entries were independently verified against publisher, preprint-server, or bibliographic-database records before inclusion; no reference was accepted without such verification. The author remains solely responsible for the mathematical claims, their attribution, and the final manuscript.

A Clarifications on Cycle Anchors, Scope, and the Multi-Window Problem

This appendix records two clarifications identified during an adversarial pre-publication audit of the manuscript. Neither changes the quantitative results proved in the body of the paper, but both sharpen the logical scope of the claims and prevent two distinct notions from being conflated.

A.1 Clarification of the anchor-cycle correspondence

Proposition 7.1 associates to a valuation block $X = (x_1, \dots, x_q)$ the rational 2-adic anchor

$$\xi_X = -\frac{C(X)}{3^q - 2^{S_X}}.$$

The cyclic affine identity shows that

$$\xi_X \in \mathbb{Z}_{>0}$$

if and only if there exists a genuine positive accelerated Collatz cycle whose valuation word, when read beginning at the state ξ_X , is exactly X .

The correct conclusion is therefore that ξ_X is *an element* of the corresponding cycle, namely the uniquely determined positive integer at which the cyclic word begins with the specified phase X . The argument given in Proposition 7.1 does not by itself compare this state with the states obtained from cyclic rotations of X , and hence does not establish that ξ_X is the numerically smallest element of the cycle.

Accordingly, any wording in the main text describing ξ_X as “the minimum element” or “the unique minimum” should be read in the following corrected form:

ξ_X is a positive integer if and only if it is an element of a genuine positive accelerated Collatz cycle whose valuation word, beginning at that element, is exactly X ; for the fixed phase X , the corresponding positive integer solution is unique.

This clarification does not affect any downstream theorem in the present paper. The later uses of Proposition 7.1 require only the implication

$$\xi_X \in \mathbb{Z}_{>0} \implies X \text{ is realized by a genuine positive accelerated cycle,}$$

not any minimality statement among cyclic rotations.

A.2 Single-window versus multi-window scope

The launch-renormalization results of this paper concern a *single-window* confinement problem. A word D_N is assumed to satisfy

$$R_j \leq c \quad (0 \leq j \leq N),$$

throughout one contiguous initial segment, and the launch-state collapse then identifies a suffix of that same segment with the accelerated orbit of a smaller integer.

Thus launch renormalization acts schematically as

one confined word $\mapsto$ a shorter confined orbit segment of the same type.

Its iterates remain within this single-window problem class, although the normalization and effective corridor width change under the reduction.

This is distinct from the multi-window occupation quantity

$$O_c(m_0) = \#\{j \geq 0 : R_j(m_0) \leq c\},$$

which counts all visits of the orbit to the region $R_j \leq c$, including later returns after excursions above the threshold. A trajectory may have the form

confined interval $\rightarrow$ excursion above $c \rightarrow$ later confined interval $\rightarrow \dots$,

and none of the launch-renormalization theorems proved here bounds the number of such separated return windows.

Consequently, this paper does *not* prove a bound of the form

$$O_c(m_0) = O(\log m_0),$$

does not prove the full multi-window Effective Occupation Conjecture, and does not close the single-window/multi-window distinction discussed in Proposition 5.10 of the companion EOC note.

What is established here is instead a structural decomposition of the single-window least-realizer problem: an anomalously small realizer yields a long forced suffix; bounded-period periodic and trivial-cycle tails admit specific obstructions; and the residual irregular class regenerates a problem of the same single-window form under launch renormalization.

A.3 What would be required to bridge to occupation counts

A proof of the full occupation bound would require genuinely new control over *returns*, not merely over the duration of an individual confined segment. In particular, a theorem bounding the length of every confined window separately does not by itself bound

$$O_c(m_0),$$

because an orbit could in principle contain many short confined windows separated by excursions outside the corridor.

A sufficient future mechanism would therefore have to control, directly or indirectly, the cumulative contribution of all such windows. This could take the form of a renewal bound, a summable per-return resource, or another quantity that prevents arbitrarily many re-entries. No such theorem is proved in the present manuscript, and no claim to one is intended.

A.4 Relation to the conditional lifetime hypotheses

The hypotheses $U_{\text{all}}(K)$ and $U_{\text{half}}(K, q_0)$ introduced in Definition 11.2 are likewise single-window statements: they bound the length of an injective initial orbit segment satisfying a confinement condition over that segment. Their role in Proposition 11.3 is to close the residual *single-window* branch conditionally and produce an exponential least-realizer floor.

They should not be interpreted as occupation-count hypotheses and do not, without an additional return-counting mechanism, imply a bound on the number of disjoint confined windows along a full orbit.

A.5 Status after clarification

With these corrections, the logical status of the manuscript is:

- the launch-state collapse and forced-continuation theorem remain unchanged;
- the terminal-suffix, periodic-tail, cycle-tail, corridor-scaling, and renormalization results remain unchanged;
- the quantitative constants $\alpha\varepsilon$, $\theta^*(\varepsilon)$, $2\alpha - 1$, and $\varepsilon_0(K)$ are unaffected;
- the anchor-cycle statement is weakened only by removing an unproved numerical-minimality clause;
- the paper is explicitly scoped to the single-window least-realizer/confinement problem and makes no claim to establish the multi-window occupation bound.

These clarifications are therefore matters of logical precision and scope, not revisions of the paper’s central quantitative chain.

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